

SocialWise: LLM-Agentic Conversation Therapy for Individuals with Autism Spectrum Disorder to Enhance Communication Skills

Albert Tang

Marriotts Ridge High School
Marriottsville, Maryland, USA
alberttang2005@gmail.com

Abstract

Autism Spectrum Disorder (ASD) affects more than 75 million people worldwide. However, scalable support for practicing everyday conversation is scarce: Low-cost activities such as story reading yield limited improvement. At the same time, effective role-play therapy demands expensive, in-person sessions with specialists. SocialWise bridges this gap through a browser-based application that pairs LLM conversational agents with a therapeutic retrieval augmented generation (RAG) knowledge base. Users select a scenario (e.g., ordering food, joining a group), interact by text or voice, and receive instant, structured feedback on tone, engagement, and alternative phrasing. The SocialWise prototype, implemented with Streamlit, LangChain, and ChromaDB, runs on any computer with internet access, and demonstrates how recent advances in LLM can provide evidence-based, on-demand communication coaching for individuals with ASD.

Keywords

SocialWise; LLM; Agents; Conversation Therapy; Autism Spectrum Disorder

ACM Reference Format:

Albert Tang. 2025. SocialWise: LLM-Agentic Conversation Therapy for Individuals with Autism Spectrum Disorder to Enhance Communication Skills. In *Proceedings of the 27th International Conference on Multimodal Interaction (ICMI '25)*, October 13–17, 2025, Canberra, ACT, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3716553.3757091>

1 Introduction

Autism Spectrum Disorder (ASD) affects 1 in 36 children in the United States and more than 75 million people worldwide [6]. About 40 % of autistic adults have never held paid employment [9].

Limits of current support. Speech therapy improves turn-taking and topic maintenance [8], but these services remain constrained by cost, geography, and provider availability [1]. Low-cost activities such as story reading yield only modest gains [5], while augmented-reality tools still require therapist supervision [3]. A scalable alternative must deliver realistic role-play and personalized feedback without specialist oversight.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

ICMI '25, Canberra, ACT, Australia

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1499-3/25/10

<https://doi.org/10.1145/3716553.3757091>

Our solution. SocialWise is a browser-based application that pairs LLM agents with a retrieval-augmented knowledge base. Users choose a scenario (e.g., ordering food), interact via text or voice (OpenAI Whisper STT + TTS), and receive instant, structured feedback tailored to ASD communication goals.

Demo contributions.

- A multimodal web interface that delivers therapist-style role-play for individuals with ASD anytime, anywhere.
- An LLM-driven feedback agent grounded by curated therapeutic documents.
- Preliminary user data (N = 34) indicating high perceived helpfulness (4.15 / 5) and 100 % willingness to recommend.

In a live demonstration, visitors will experience a full interaction cycle—scenario selection, live conversation, and feedback—in under three minutes, demonstrating how recent LLM advances can make evidence-informed communication coaching broadly accessible.

2 Background & Related Work

To position SocialWise, we review prior efforts along four aspects: traditional role-play interventions, digital communication aides, LLM role-play agents, and educational chatbots.

Role-play interventions. Classical Social Skills Training (SST) programs combine modelling, role-play, and feedback, and remain the gold standard for pragmatic language gains in ASD [4]. Cost and geography, however, restrict access [1].

Digital aides. Augmentative-and-Alternative Communication (AAC) tools [7] and AR systems such as STAR [3] broaden reach yet still require therapist supervision and provide little adaptive feedback.

LLM role-play agents. Character-centric frameworks (e.g., RoleLLM [10] and CharacterGLM [12]) can sustain persona-consistent dialogue but do not target therapeutic coaching.

Educational chatbots. AI tutors improve skill rehearsal in translation [11] and interviewing [2], yet lack grounding in ASD communication goals.

Closing the remaining gaps. SocialWise delivers (1) **adaptive feedback** by coupling GPT-4o-mini dialogue with a retrieval-grounded analysis engine aligned to evidence-based ASD therapy guidelines; (2) seamless **speech & text interaction** via Whisper STT and OpenAI TTS, enabling fully multimodal practice; and (3) true **therapist-free accessibility**, as the entire system runs in a standard web browser without specialised hardware or professional supervision.

3 System Snapshot

Architecture. SocialWise is a Streamlit web app that calls GPT-4o-mini for dialogue, Whisper STT, and OpenAI TTS. A LangChain

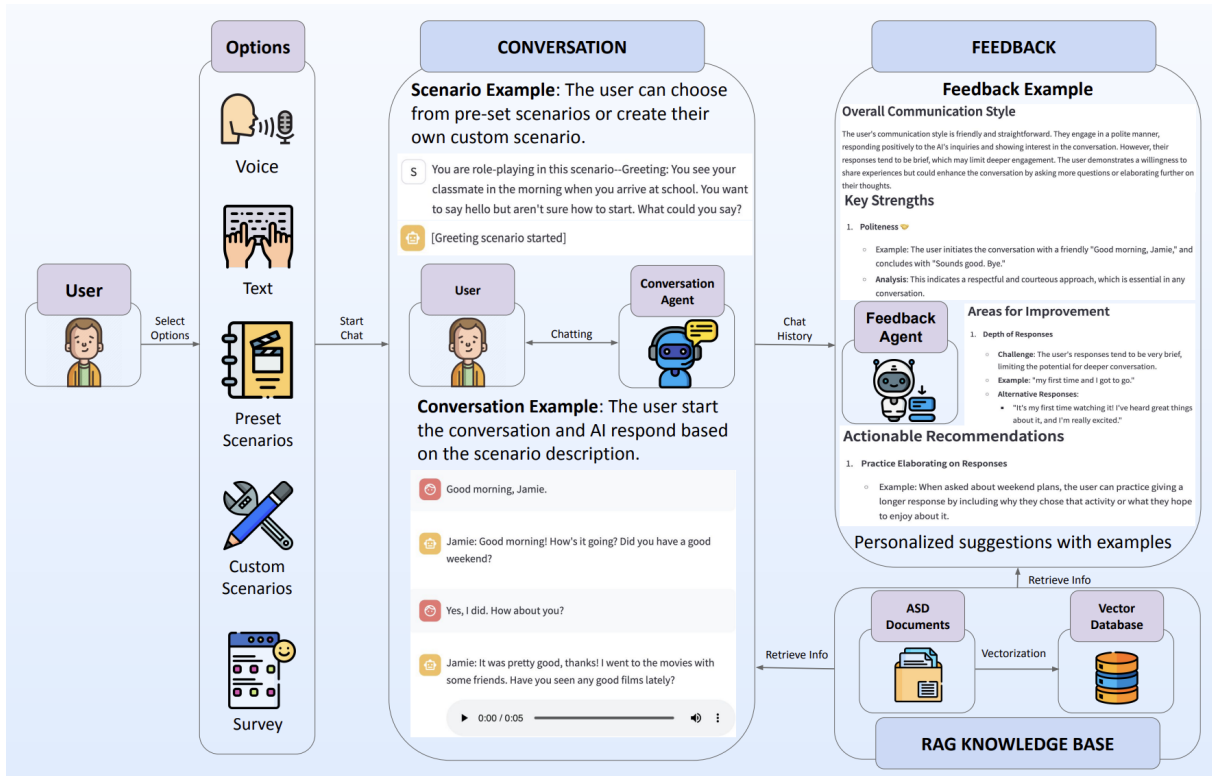


Figure 1: The workflow of SocialWise, an AI-powered conversation-therapy chatbot that helps individuals with autism practise social interactions. The user selects conversation options, role-plays with a conversation agent, and the system retrieves relevant knowledge from a RAG-powered vector database. A feedback agent analyses user responses and provides constructive feedback.

Retrieval-Augmented Generation pipeline (ChromaDB dense retriever) grounds feedback in ASD-therapy documents.

Runtime. Average latency is 1.6s for text and 4.8s for speech on a 2021 M1 laptop; Our system is scalable because of the use of OpenAI's API and Streamlit's deployment system, and we will use FastAPI with NextJS for future production.

Safety. Hallucination is reduced by RAG grounding, instruction prompts, and expert review; CoT bias-mitigation prompts are enabled, with fine-tuning planned.

4 Demo Interaction Flow

Figure 2 shows that a user's journey involves four major steps:

- (1) *Scenario pick.* Visitors select a preset role-play scenario (e.g. "Ordering food") or type their own situation.
- (2) *Live role-play.* They speak or type; GPT-4o-mini answers in natural speech and on-screen text while retaining context.
- (3) *Instant feedback.* Pressing GET FEEDBACK triggers retrieval-grounded analysis that appears in under three seconds, highlighting strengths, areas to improve, and alternative phrasings.
- (4) *Iterate / restart.* Attendees can continue the conversation, switch scenarios, or export the feedback as HTML.

Figure 2 shows an annotated screenshot of the SocialWise web interface. The left sidebar (*Options*) lets users toggle text-to-speech and speech-to-text, pick a pre-set scenario, or type a custom scenario; a link at the bottom opens the user-feedback survey. The centre column displays (top-to-bottom) an expandable "How to Use" panel, the scenario description, the live role-play history, an optional audio-playback bar, buttons to restart the chat or request feedback, and a text-input field. The right column presents the structured feedback: overall style, key strengths, areas for improvement, and actionable recommendations, with buttons below to download the analysis or return to the conversation. The feedback is on a separate page on the web app.

The layout makes all critical functions—scenario selection, multimodal conversation, and evidence-based feedback for the live demo.

5 Preliminary User Feedback

We ran a remote pilot with **34 autistic adults** (ages 18–34). After completing at least one role-play, participants rated overall helpfulness **4.15 / 5** and **100 %** said they would recommend the tool to a peer. The three most cited benefits were improved *turn-taking* (67.6 %), *problem-solving in social conflicts* (52.9 %), and *reduced anxiety during real conversations* (50.0 %). Open-ended suggestions focused

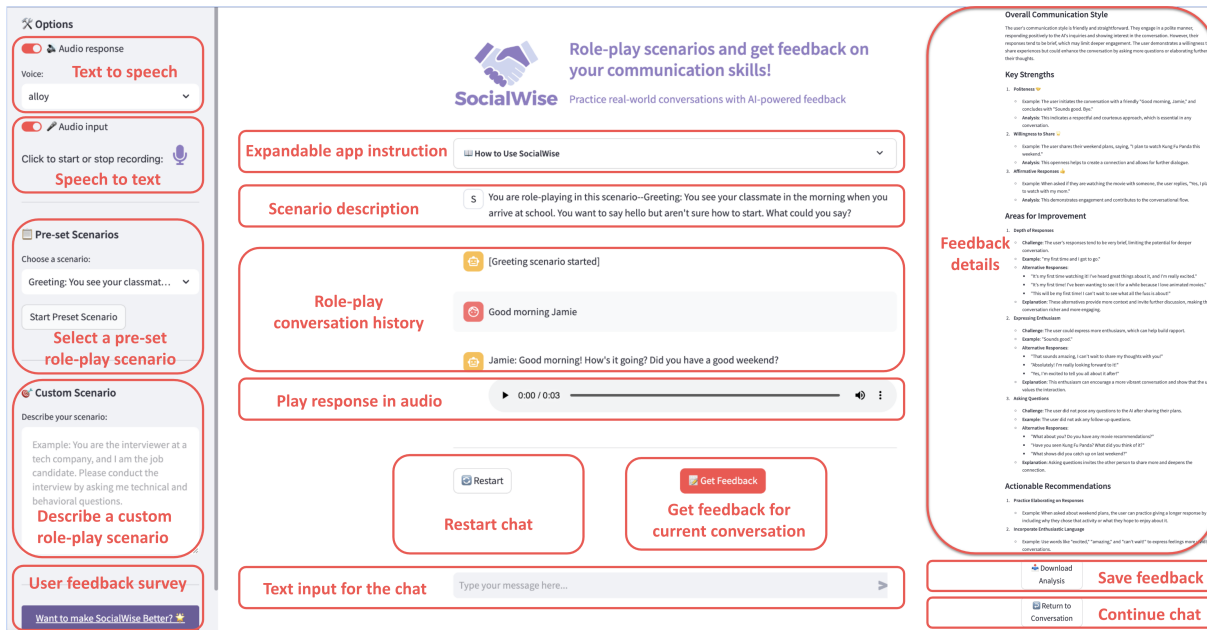


Figure 2: Annotated screenshot of the SocialWise web interface.

on adding gamified rewards and advanced scenarios, confirming user appetite for deeper practice. Although not a clinical trial, these early results indicate strong perceived value and engagement.

6 Future Work

Future work for SocialWise includes conducting formal clinical validation studies, expanding the scenario library for broader social contexts, developing a mobile app, and implementing user progress tracking features.

7 Responsible Innovation & Ethics

SocialWise is designed for safe, private, and equitable use.

Privacy. All conversation logs remain in the browser’s memory and are flushed on page refresh; no user identifiers or audio recordings are stored server-side.

Bias mitigation. We employ chain-of-thought (CoT) bias-reduction prompts at every generation step and plan a follow-up fine-tuning model that addresses community feedback.

Hallucination control. Retrieval grounding, strict instruction prompts, and human spot-checks limit factual drift.

We are committed to the responsible development and deployment of SocialWise. This includes ongoing monitoring for potential negative impacts and a commitment to transparency with our users about the capabilities and limitations of the AI.

References

- [1] Lauren Brookman-Frazer, Mary Baker-Ericzen, Nicole Stadnick, and Rachel Taylor. 2012. Parent Perspectives on Community Mental Health Services for Children With Autism Spectrum Disorders. *Journal of Child and Family Studies* 21, 4 (2012), 533–544. doi:10.1007/s10826-011-9516-8
- [2] L. Chen, V. Feng, and H. Zheng. 2021. AI-driven interview simulation for skill development. In *Proceedings of the 2021 CHI Conference on Human Factors in*

- Computing Systems (CHI ’21)*. Association for Computing Machinery (ACM), 1–12.
- [3] Camilla Almeida da Silva, António Ramires Fernandes, and Ana Paula Grohmann. 2015. STAR: Speech Therapy with Augmented Reality for Children with Autism Spectrum Disorders. In *Enterprise Information Systems. Lecture Notes in Business Information Processing*, Vol. 227. Springer International Publishing, 379–396. doi:10.1007/978-3-319-22348-3_21
- [4] Wiam El-Kassem, Hadeel Mohammad, Laila Al-Ayadhi, and Abdullah Al-Joudi. 2024. Effectiveness of social skills training interventions for children with autism spectrum disorder: A systematic review and meta-analysis. *PLOS One* 19, 3 (Mar 2024), e0299803. doi:10.1371/journal.pone.0299803
- [5] Sue Fletcher-Watson, Alexandra Petrou, Juliet Scott-Barrett, Pamela Dicks, Catherine Graham, Anne O’Hare, Helen Pain, and Helen McConachie. 2016. A trial of an iPad™ intervention targeting social communication skills in children with autism. *Autism* 20, 7 (2016), 771–782. doi:10.1177/1362361315615031
- [6] Matthew J. Maenner, Zachary Warren, Ashley Robinson Williams, Esther Amoakohene, Amanda V. Bakian, Deborah A. Bilder, Maureen S. Durkin, Robert T. Fitzgerald, Sarah M. Furnier, Michelle M. Hughes, Christine M. Ladd-Acosta, Dedria McArthur, Elise T. Pas, Angelica Salinas, Alison Vehorn, Susan Williams, Amy Esler, Andrea Grzybowski, Jennifer Hall-Lande, Ruby H.N. Nguyen, Karen Pierce, Walter Zahorodny, Allison Hudson, Libby Hallas, Kristen Clancy Mancilla, Mary Patrick, Josephine Shenouda, Kate Sidwell, Monica DiRienzo, Johanna Gutierrez, Margaret H. Spivey, Maya Lopez, Sydney Pettygrove, Yvette D. Schwenk, Anita Washington, and Kelly A. Shaw. 2023. Prevalence and Characteristics of Autism Spectrum Disorder Among Children Aged 8 Years — Autism and Developmental Disabilities Monitoring Network, 11 Sites, United States, 2020. *MMWR Surveillance Summaries* 72, 2 (2023), 1–14. doi:10.15585/mmwr.ss7202a1
- [7] Pat Miranda. 2003. Toward Functional Augmentative and Alternative Communication for Students With Autism: Manual Signs, Graphic Symbols, and Voice Output Communication Aids. *Language, Speech, and Hearing Services in Schools* 34, 3 (2003), 203–216. doi:10.1044/0161-1461(2003/017)
- [8] Lauren Parsons, Reinie Cordier, Natalie Munro, Annette Joosten, and Renée Speyer. 2017. A systematic review of pragmatic language interventions for children with autism spectrum disorder. *PLOS One* 12, 4 (2017), e0172242. doi:10.1371/journal.pone.0172242
- [9] Paul T. Shattuck, Sarah Carter Narendorf, Benjamin P. Cooper, Paul R. Sterzing, Mary M. Wagner, and Julie Lounds Taylor. 2012. Postsecondary Education and Employment Among Youth With an Autism Spectrum Disorder. *Pediatrics* 129, 6 (2012), 1042–1049. doi:10.1542/peds.2011-2864
- [10] Zekun Moore Wang, Zhongyuan Peng, Haoran Que, Jiaheng Liu, Wangchunshu Zhou, Yuhuan Wu, Hongcheng Guo, Ruitong Gan, Zehao Ni, Man Zhang, Yaxi Xiao, Zhaoyu Li, Jushuo Ti, Zhaoxiang Zhang, Songyang Zhang, Wei

- He, Junlin Yao, Jing Chen, and Weiping Wang. 2023. RoleLLM: Benchmarking, Eliciting, and Enhancing Role-Playing Abilities of Large Language Models. arXiv:2310.00746 [cs.CL]
- [11] Steve Woollaston, Brendan Flanagan, Patrick Ocheja, Yiling Dai, and Hiroaki Ogata. 2023. TAMMY: Supporting EFL Translation Practice With an LLM-Powered Chatbot. In *Proceedings of the 31st International Conference on Computers in Education (ICCE 2023)*. Asia-Pacific Society for Computers in Education (AP-SCE), Matsue, Japan. <https://library.apsce.net/index.php/ICCE/article/view/4901>
- [12] Jinfeng Zhou, Zhuang Chen, Dazhen Wan, Bosi Wen, Yi Song, Jifan Yu, Yongkang Huang, Pei Ke, Guanqun Bi, Libiao Peng, JiaMing Yang, Xiyao Xiao, Sahand Sabour, Xiaohan Zhang, Wenjing Hou, Yijia Zhang, Yuxiao Dong, Hongning Wang, Jie Tang, and Minlie Huang. 2024. CharacterGLM: Customizing Social Characters with Large Language Models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*. Association for Computational Linguistics, Miami, Florida.